

SSAFY

Git

Remote Repository

공통



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Remote Repository

원격 저장소

원격 저장소

원격 저장소

Remote Repository

코드와 버전 관리 이력을 온라인 상의 특정 위치에 저장하여
여러 개발자가 협업하고 코드를 공유할 수 있는 저장 공간

다양한 원격 저장소 서비스



remote

로컬 & 원격 저장소

1. GitHub 가입
 - <https://github.com/>
2. 제공된 *01_Git_설치_및_GitHub_설정.pdf* 문서 참조
3. GitHub repository 생성

로컬 & 원격 저장소



GitHub Repository



remote



Local Repository



```
git remote add origin remote_repo_url
```

로컬 저장소에 원격 저장소 추가



```
git remote add origin remote_repo_url
```

추가하는 원격 저장소 별칭

별칭을 사용해 로컬 저장소 한 개에
여러 원격 저장소를 추가할 수 있음



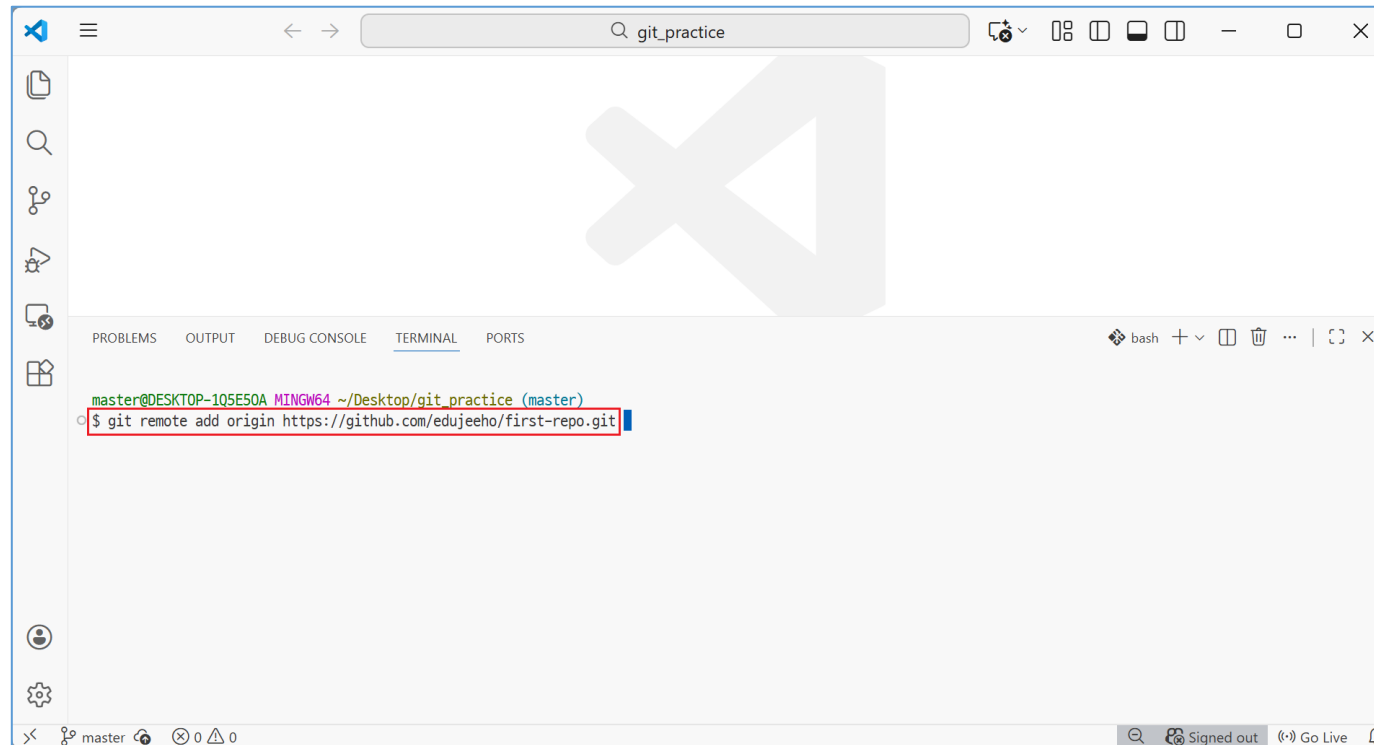
```
git remote add origin remote_repo_url
```

추가하는 원격 저장소 주소

추가하는 원격 저장소의 주소 (URL)

git remote (1/2)

- 로컬 저장소에 원격 저장소 주소 등록

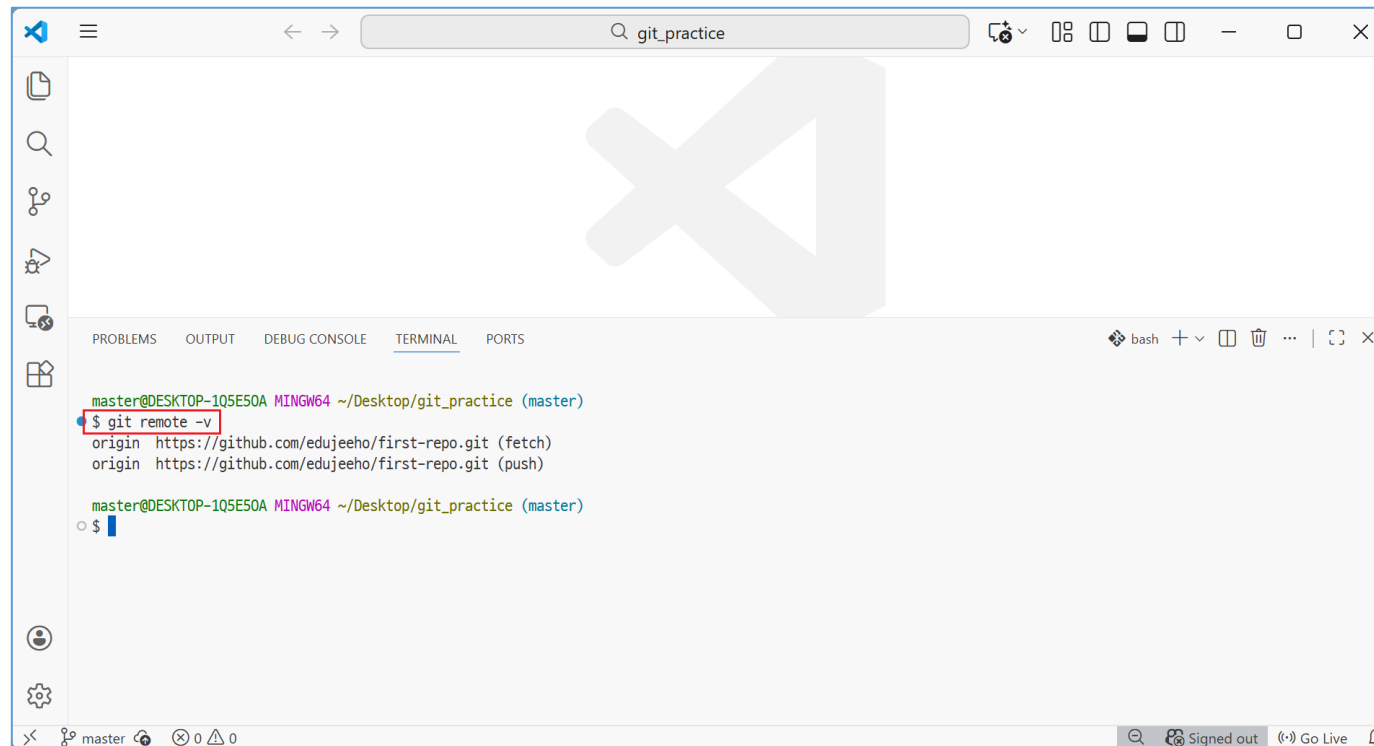


The screenshot shows a Visual Studio Code window with a terminal open. The terminal title is "git_practice". The command prompt shows the user is in a Windows environment (MINGW64) at the path ~/Desktop/git_practice, on the master branch. The command being entered is `git remote add origin https://github.com/edujeeho/first-repo.git`, which is highlighted with a red box.

```
master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_practice (master)
$ git remote add origin https://github.com/edujeeho/first-repo.git
```

git remote (2/2)

- 등록된 원격 저장소 목록 확인

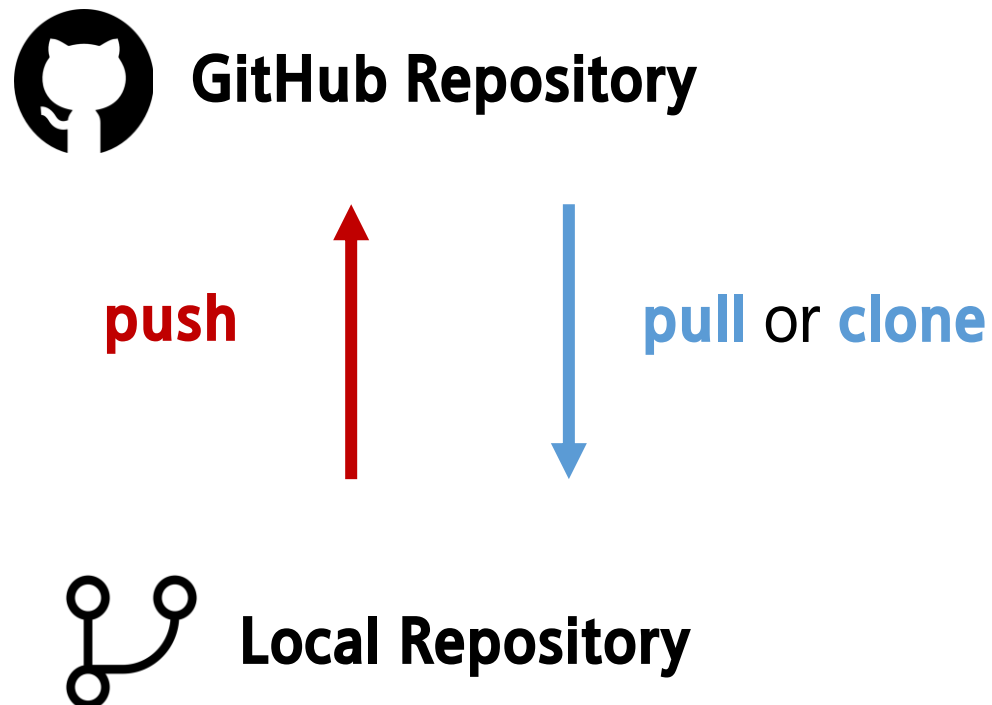


The screenshot shows a Visual Studio Code window with a terminal at the bottom. The terminal is running a Windows command prompt (MINGW64) in the directory ~/Desktop/git_practice. The user has executed the command `git remote -v`, which is highlighted with a red box. The output shows two remote repositories, both named 'origin', with the same HTTPS URL: `https://github.com/edujeeho/first-repo.git`. The first line indicates the 'fetch' operation, and the second line indicates the 'push' operation. The terminal prompt is `master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_practice (master)`.

```
master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_practice (master)
$ git remote -v
origin https://github.com/edujeeho/first-repo.git (fetch)
origin https://github.com/edujeeho/first-repo.git (push)

master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_practice (master)
$
```

push

 push / pull & clone

 push / pull & clone

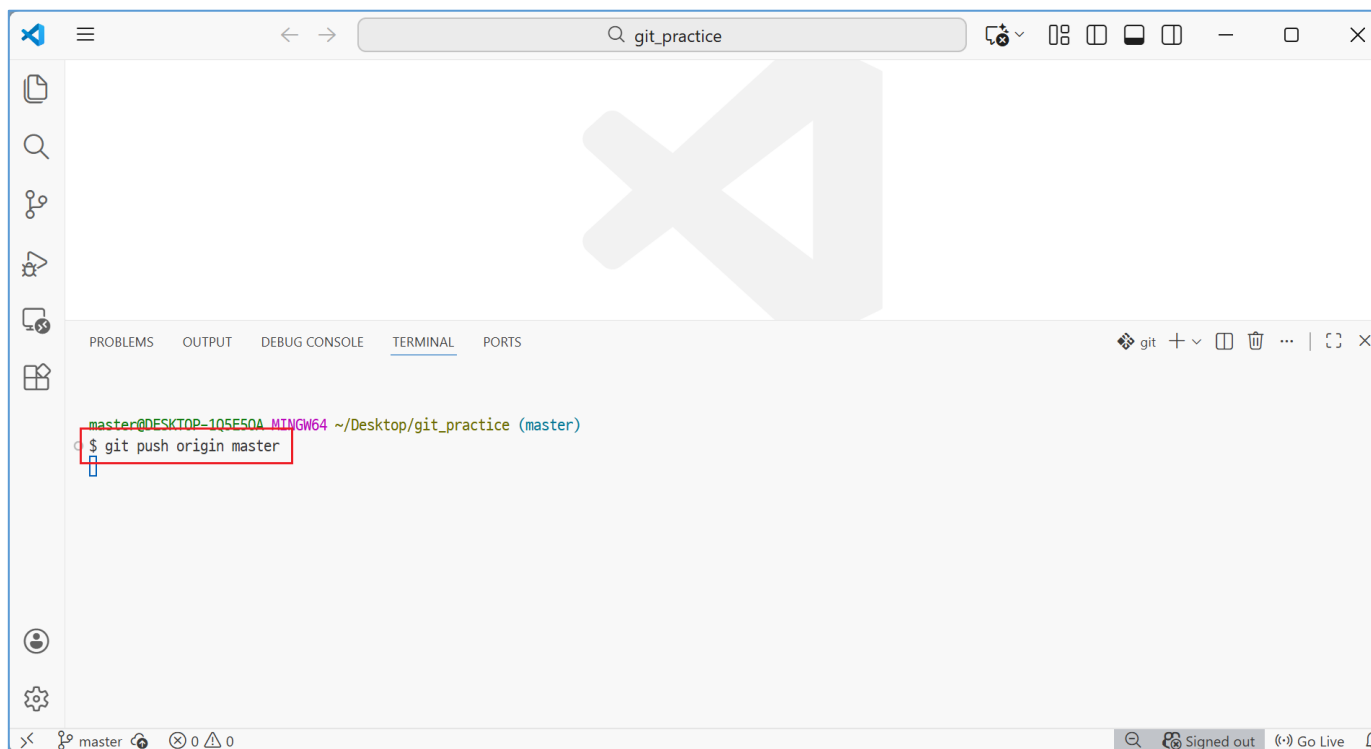
git push origin master

원격 저장소에 commit 목록을 업로드

"git아, push 해줘 origin이라는 이름의 원격 저장소에 master라는 이름의 브랜치를"라는 의미의 명령어입니다.

git push (1/5)

- commit 목록 업로드

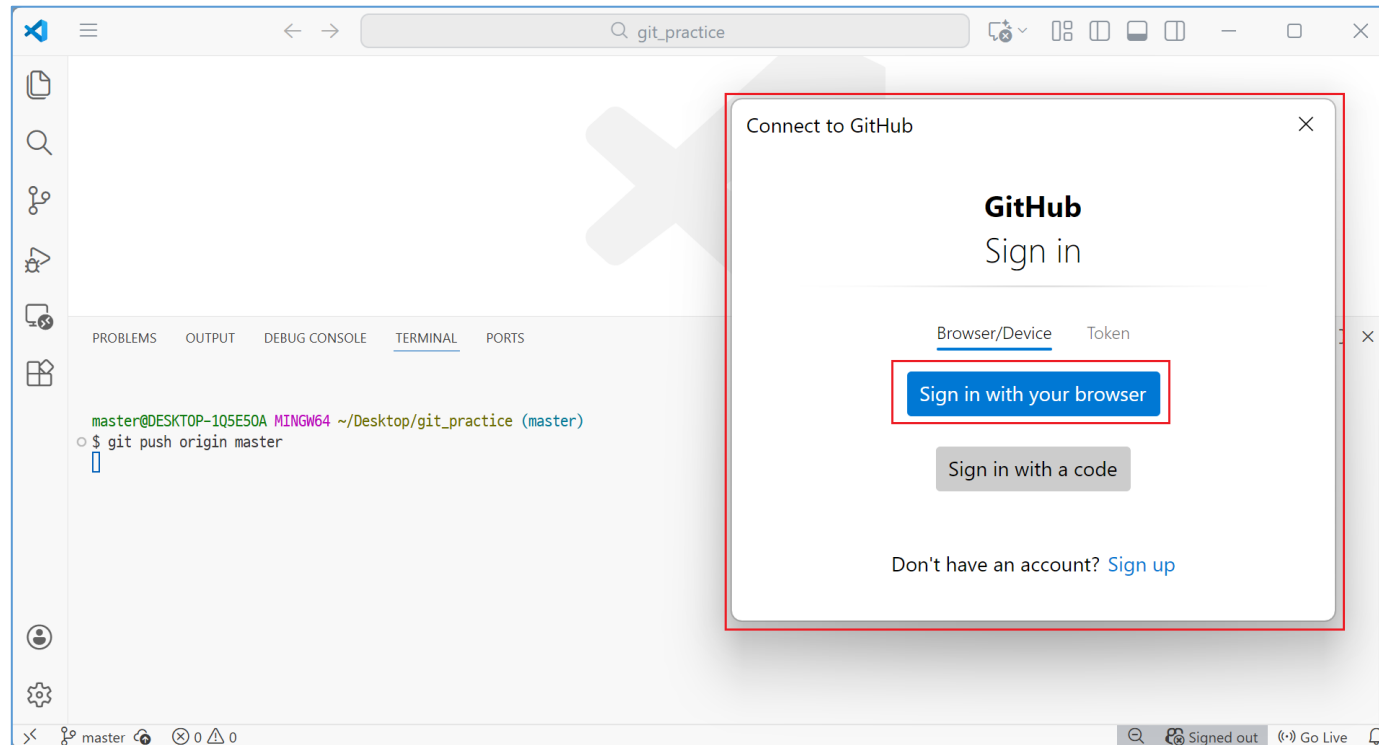


The screenshot shows a Visual Studio Code window with a terminal at the bottom. The terminal prompt is `master@DESKTOP-10SE50A: ~/Desktop/git_practice (master)`. The command `$ git push origin master` is being typed and is highlighted with a red box. The background of the terminal has a large, faint 'X' watermark.

```
master@DESKTOP-10SE50A: ~/Desktop/git_practice (master)
$ git push origin master
```

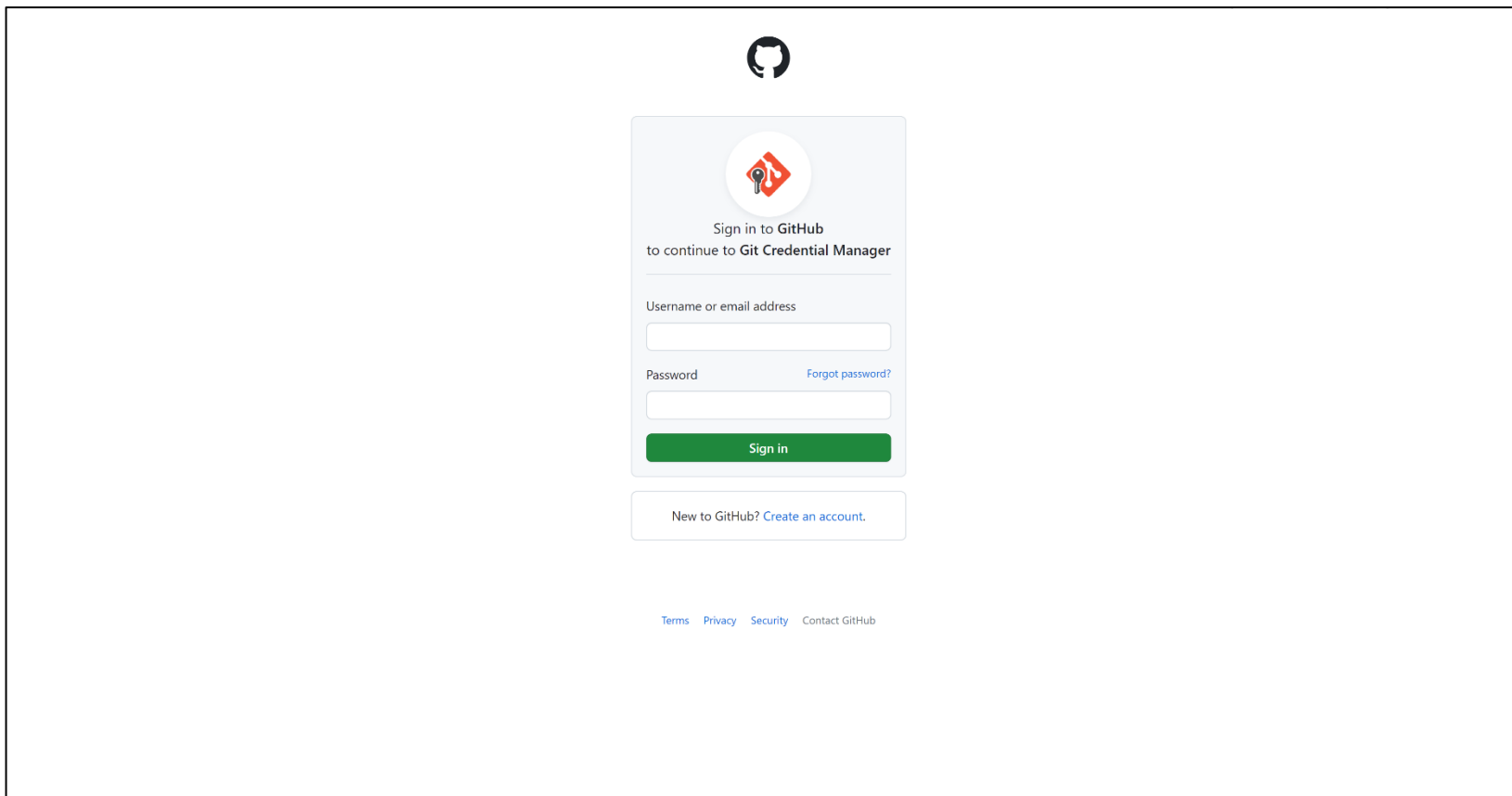
git push (2/5)

- 최초 push 시에는 GitHub으로부터 인증서(git credential) 발급이 필요
- 해당 원격 저장소에 push 할 수 있는 권한이 있는지 확인하기 위함



git push (3/5)

- GitHub 로그인



The screenshot shows the GitHub login interface. At the top center is the GitHub Octocat logo. Below it is a light gray box containing the Git Credential Manager logo (a red square with a white key icon). The text inside the box reads: "Sign in to GitHub to continue to Git Credential Manager". Below this text are two input fields: "Username or email address" and "Password". To the right of the password field is a blue link that says "Forgot password?". Below the input fields is a green button with the text "Sign in". At the bottom of the box is a link that says "New to GitHub? Create an account." Below the box, at the bottom of the page, are links for "Terms", "Privacy", "Security", and "Contact GitHub".

git push (4/5)

- 터미널 메시지를 통해 push가 성공했다는 것을 확인

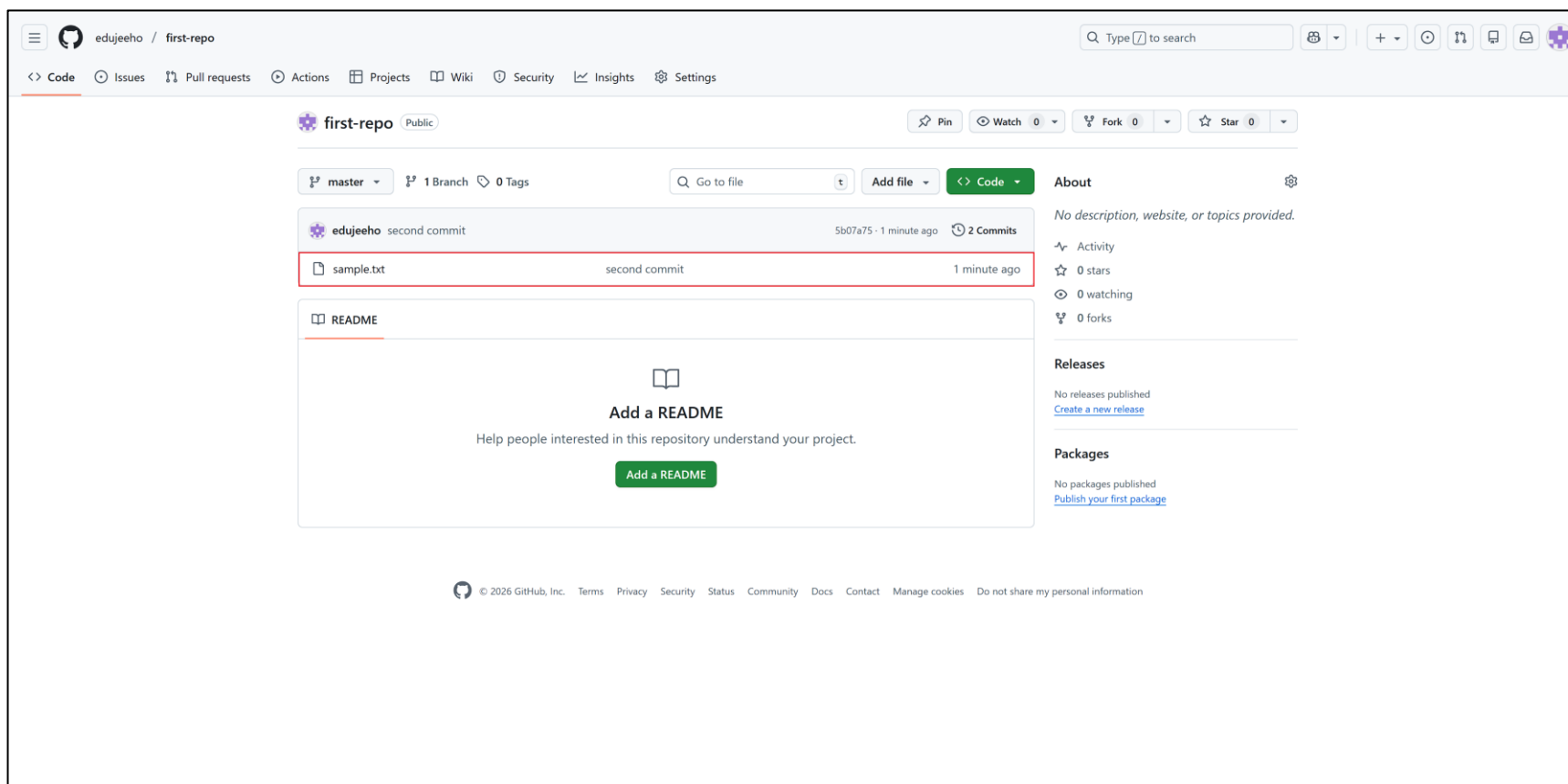


```
master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_practice (master)
$ git push origin master
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 482 bytes | 482.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/edujeeho/first-repo.git
 * [new branch]      master -> master

master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_practice (master)
$
```

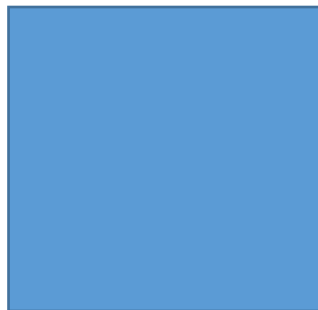
git push (5/5)

- 원격 저장소에 commit 목록이 잘 push 되었는지 확인



push 다시보기 (1/4)

- 로컬 저장소에 commit 생성



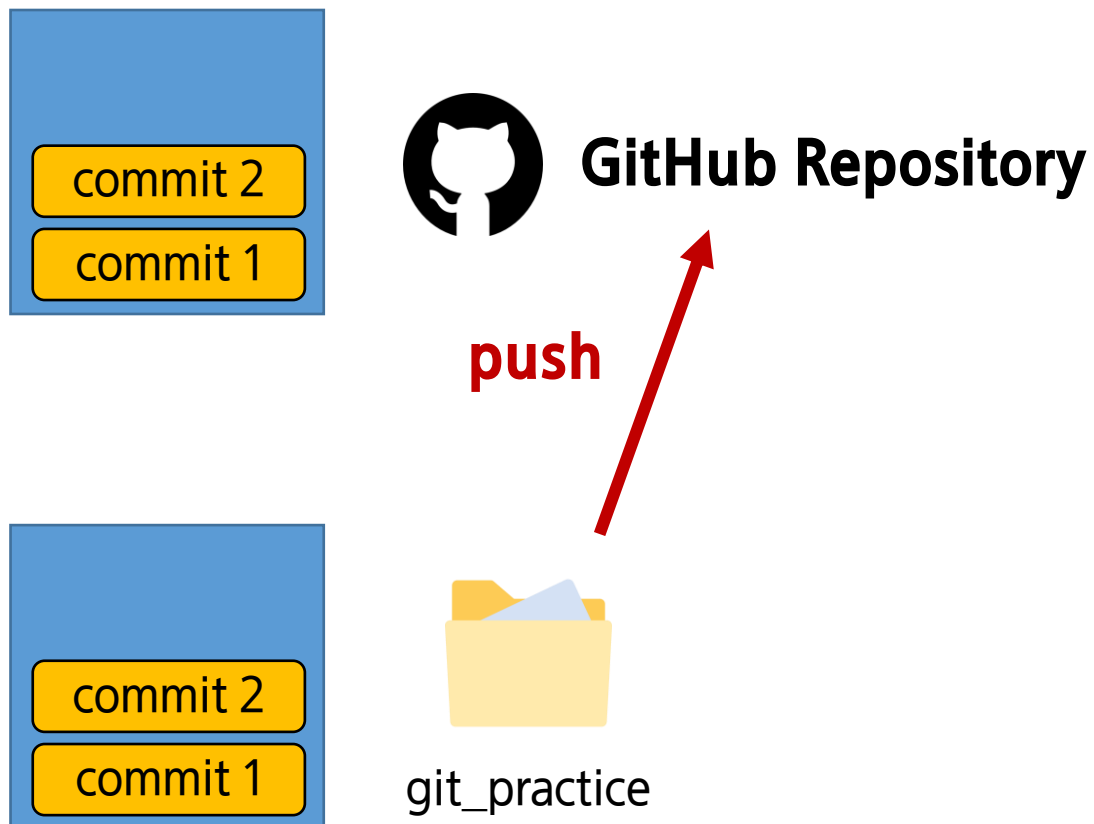
GitHub Repository



git_practice

push 다시보기 (2/4)

- 로컬 저장소의 commit을 push



push 다시보기 (3/4)

- 로컬 저장소에 3번째 commit 생성



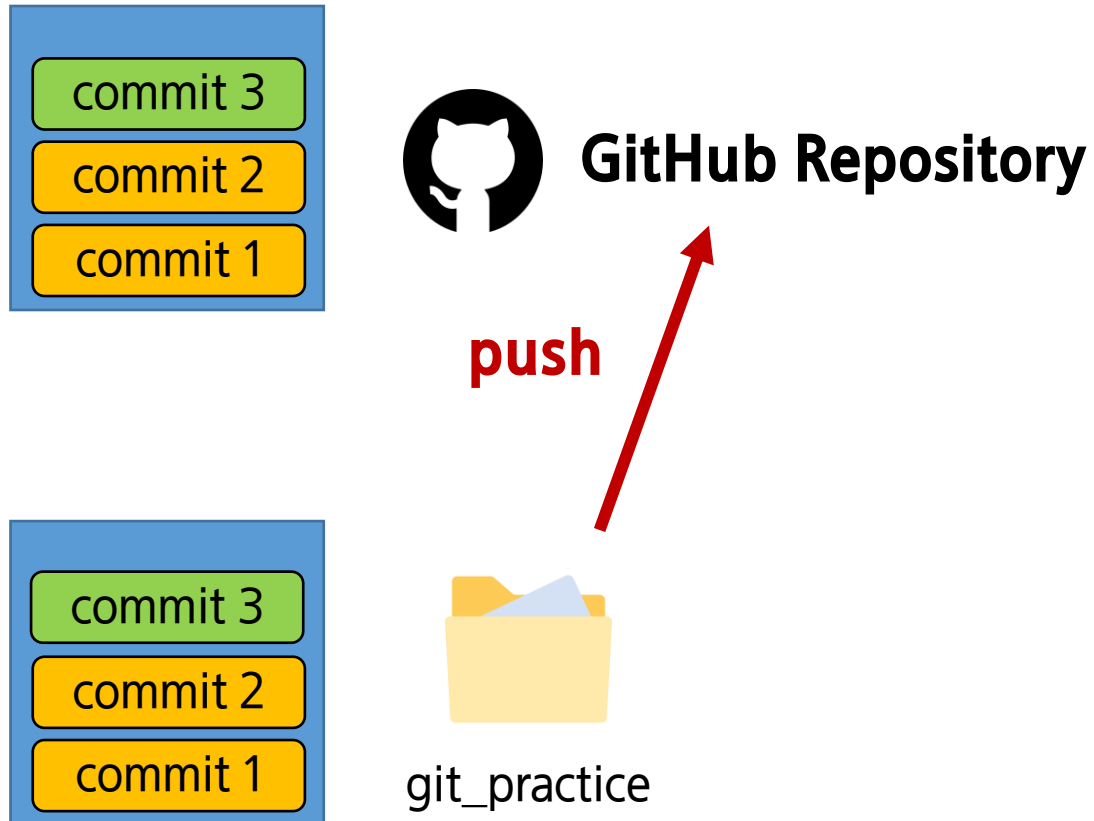
GitHub Repository



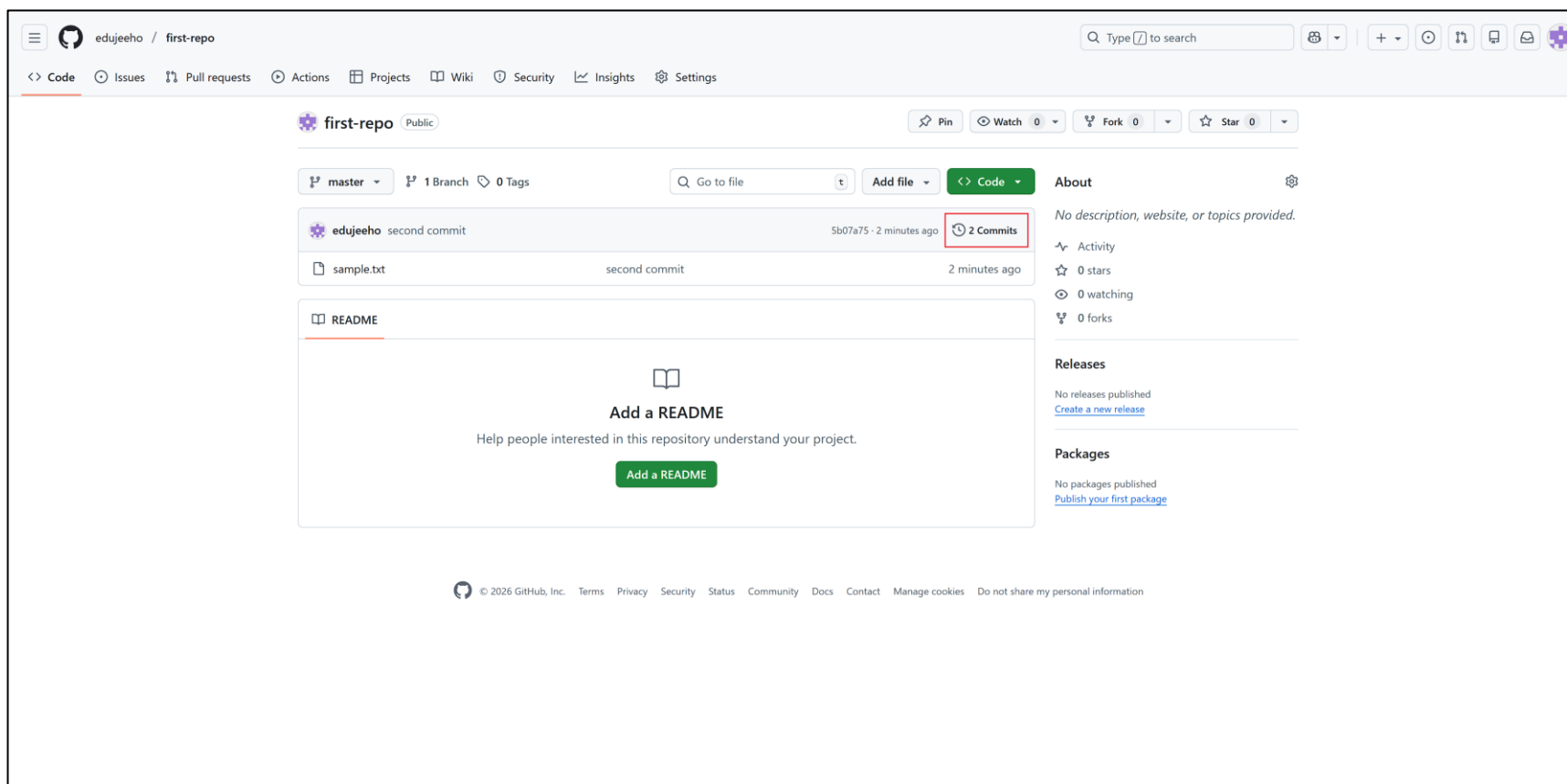
git_practice

push 다시보기 (4/4)

- 로컬 저장소의 commit을 push (변경사항 만큼 업로드)

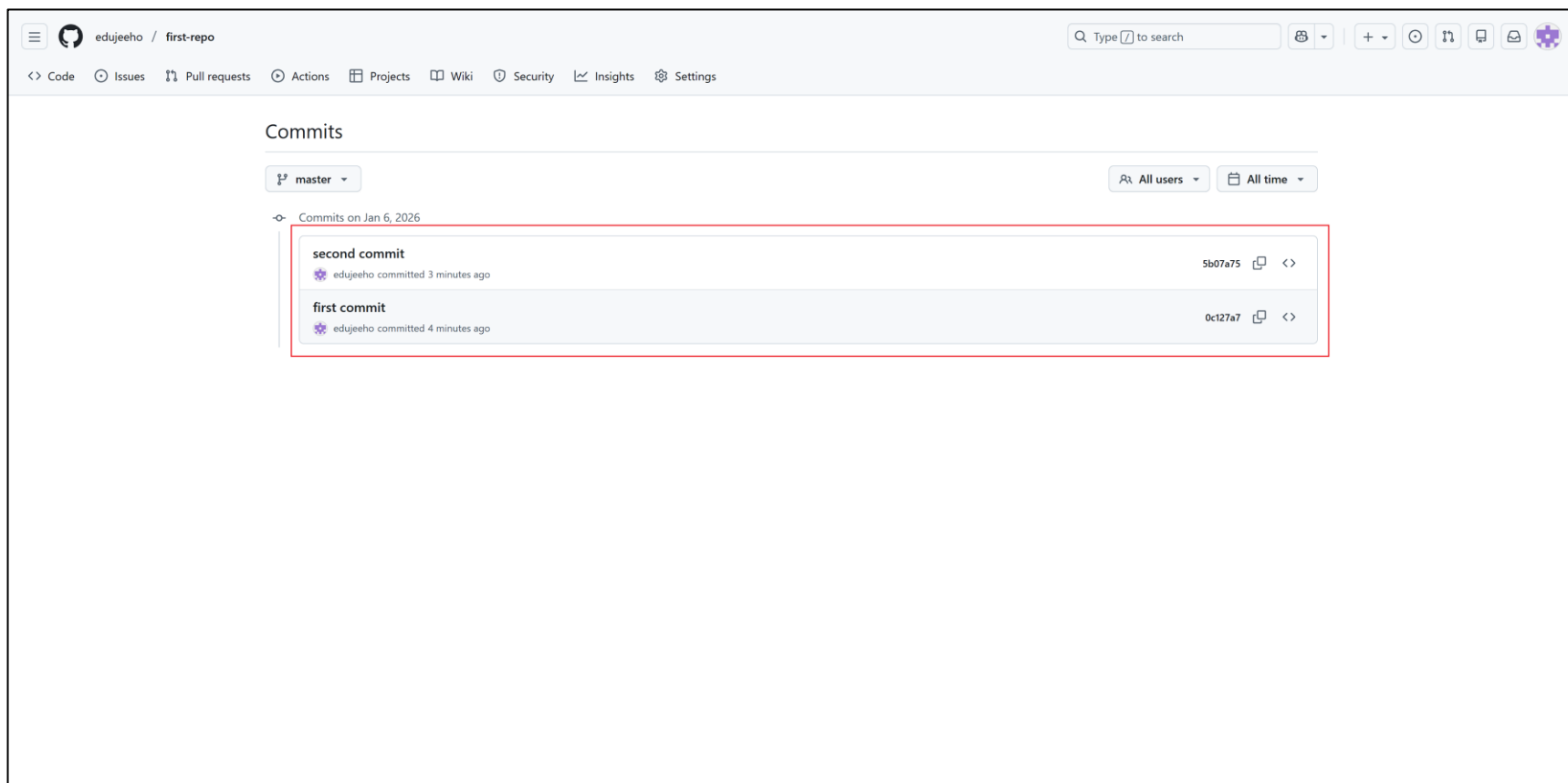


원격 저장소에서의 commit 확인 (1/3)



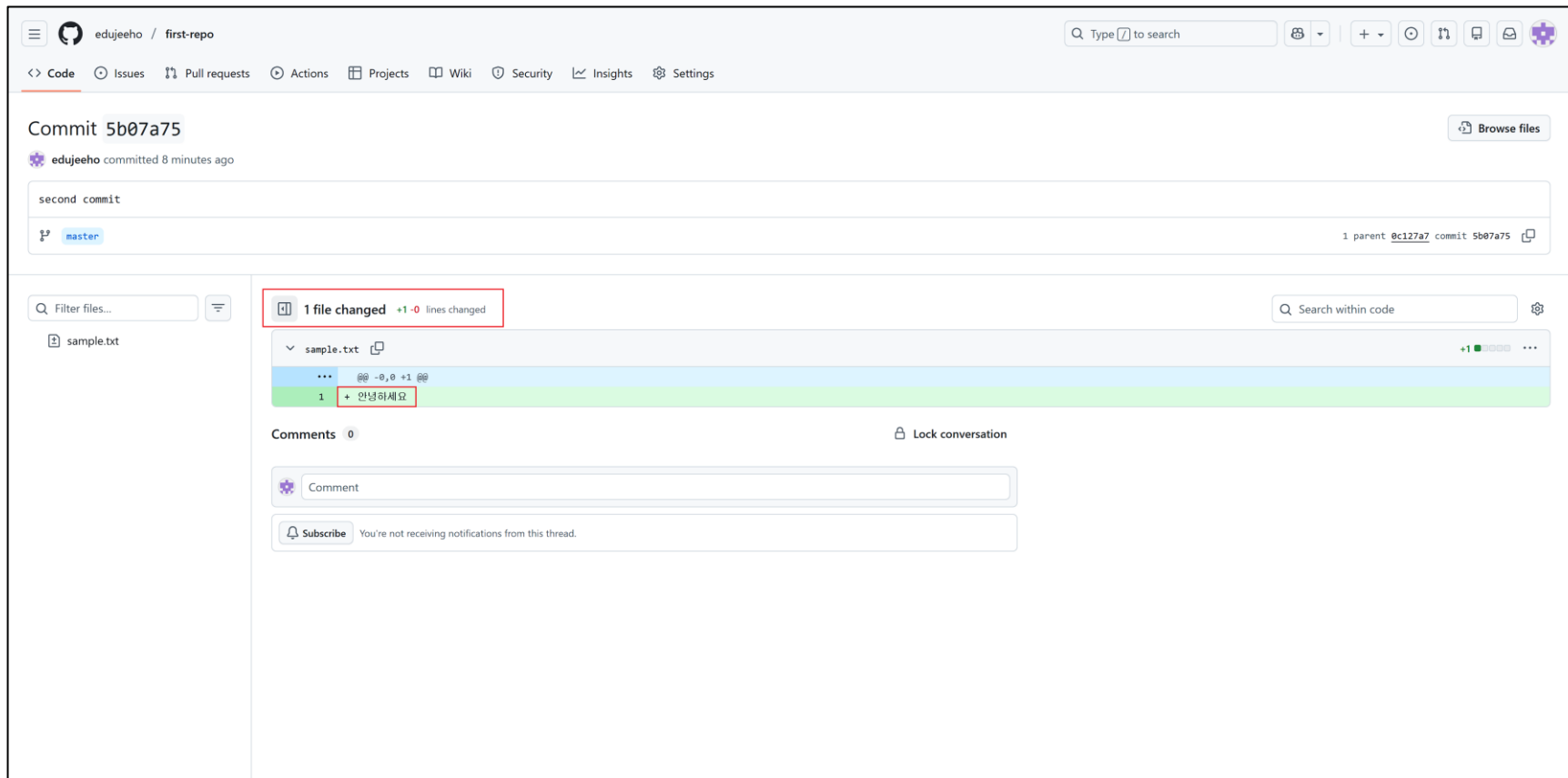
원격 저장소에서의 commit 확인 (2/3)

- 로컬 저장소에서 만들었던 commit 목록 확인



원격 저장소에서의 commit 확인 (3/3)

- commit 내역을 상세하게 보면 어떤 변경사항에 대한 commit인지 확인 가능

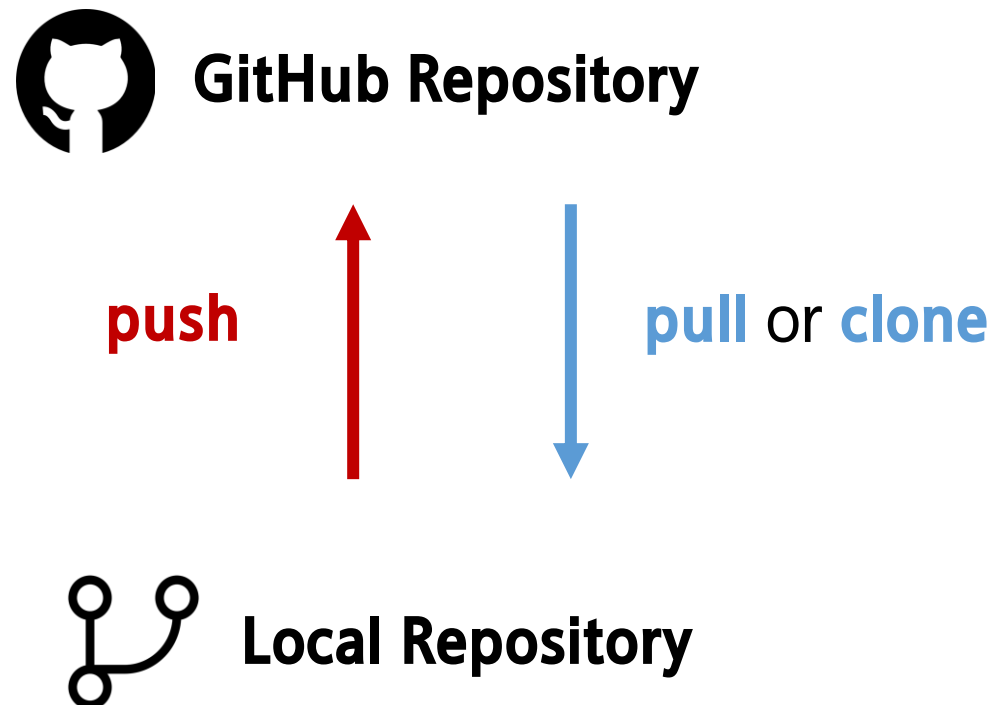


 push 결론

원격 저장소에는 commit이 올라가는 것

commit 이력이 없다면 push 할 수 없다.

pull & clone

 push / pull & clone

 push / pull & clone

```
git pull origin master
```

원격 저장소의 변경사항만을 받아옴 (업데이트)

 push / pull & clone

git clone remote_repo_url

원격 저장소 전체를 복제 (다운로드)

clone 받은 프로젝트는 이미 git init이 되어 있습니다.

git pull & clone 미리보기 (1/4)

- 2개의 폴더를 사용해 2명의 사용자가 있다고 가정하기



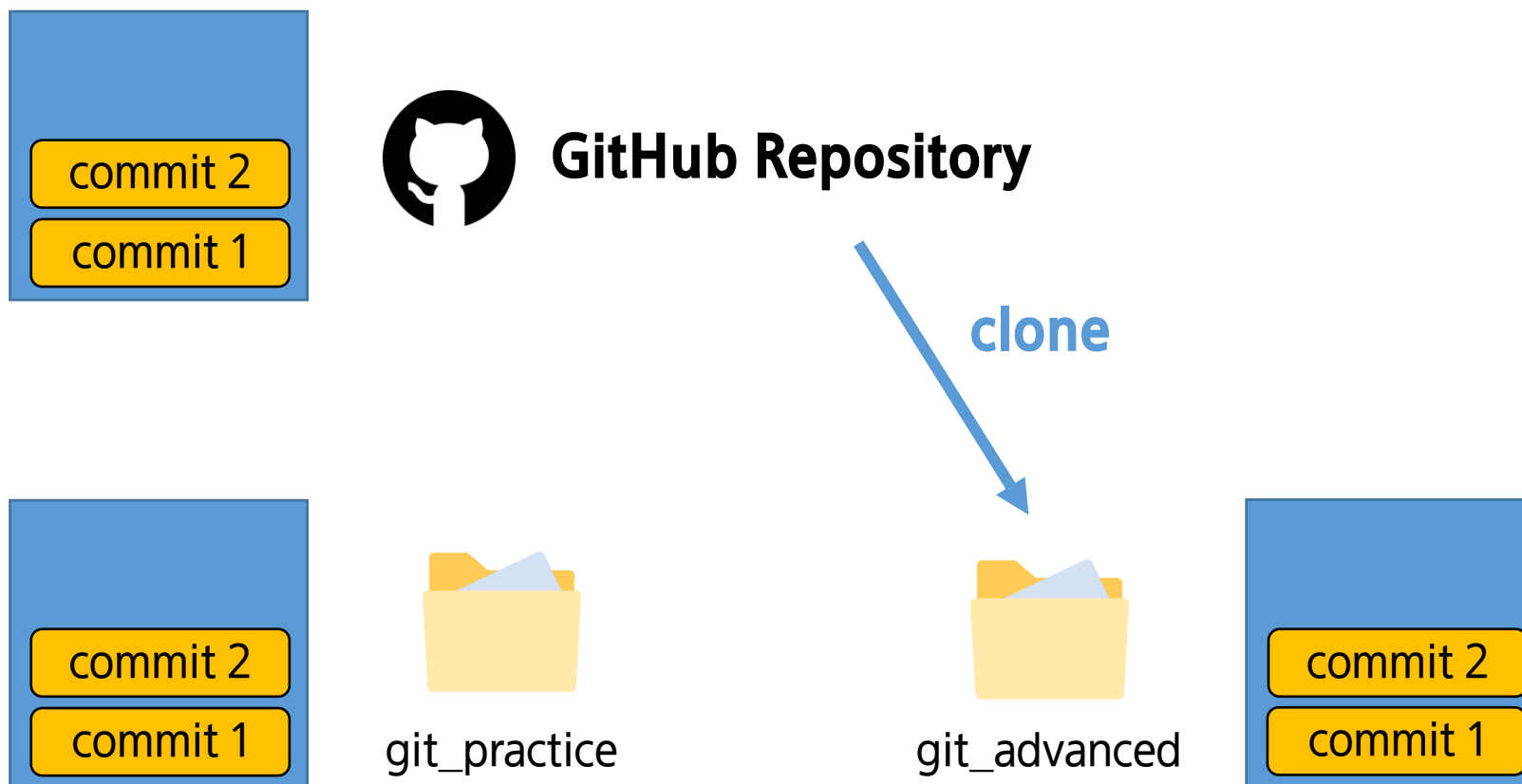
GitHub Repository



git_practice

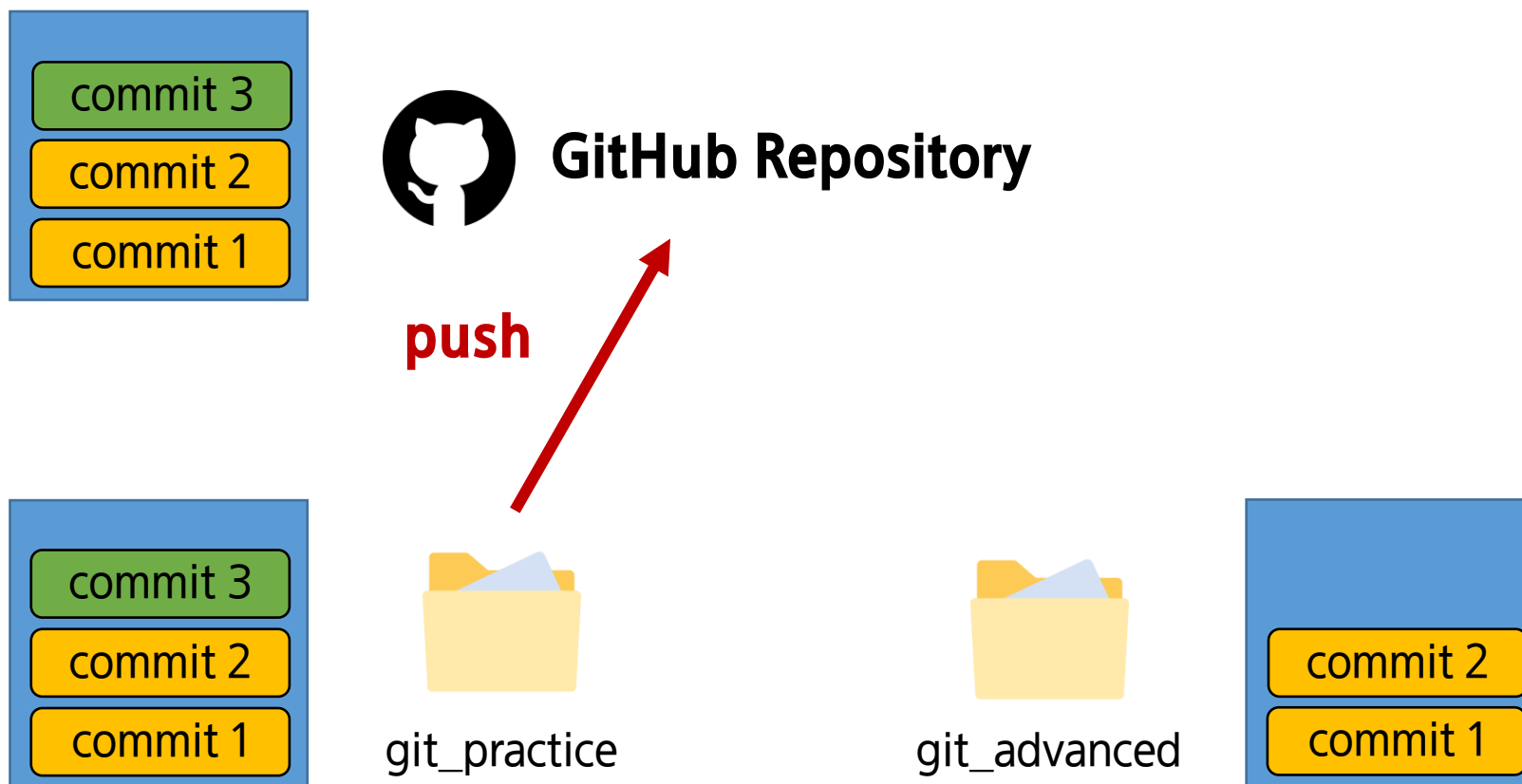
git pull & clone 미리보기 (2/4)

- 2번 유저는 해당 프로젝트를 처음 받는 것이기 때문에 저장소 전체를 복제



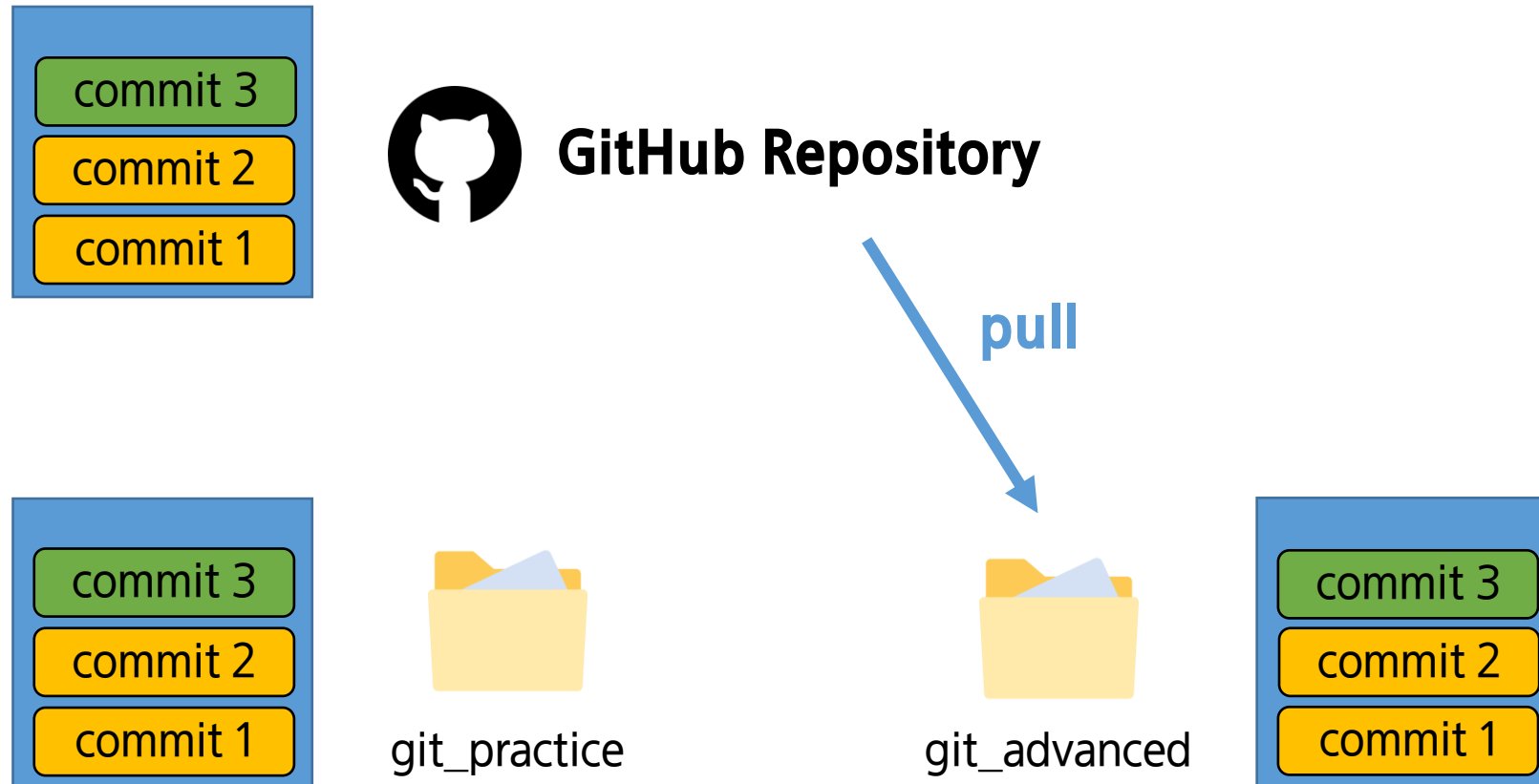
git pull & clone 미리보기 (3/4)

- 1번 유저가 추가 commit을 생성 후 원격 저장소에 push



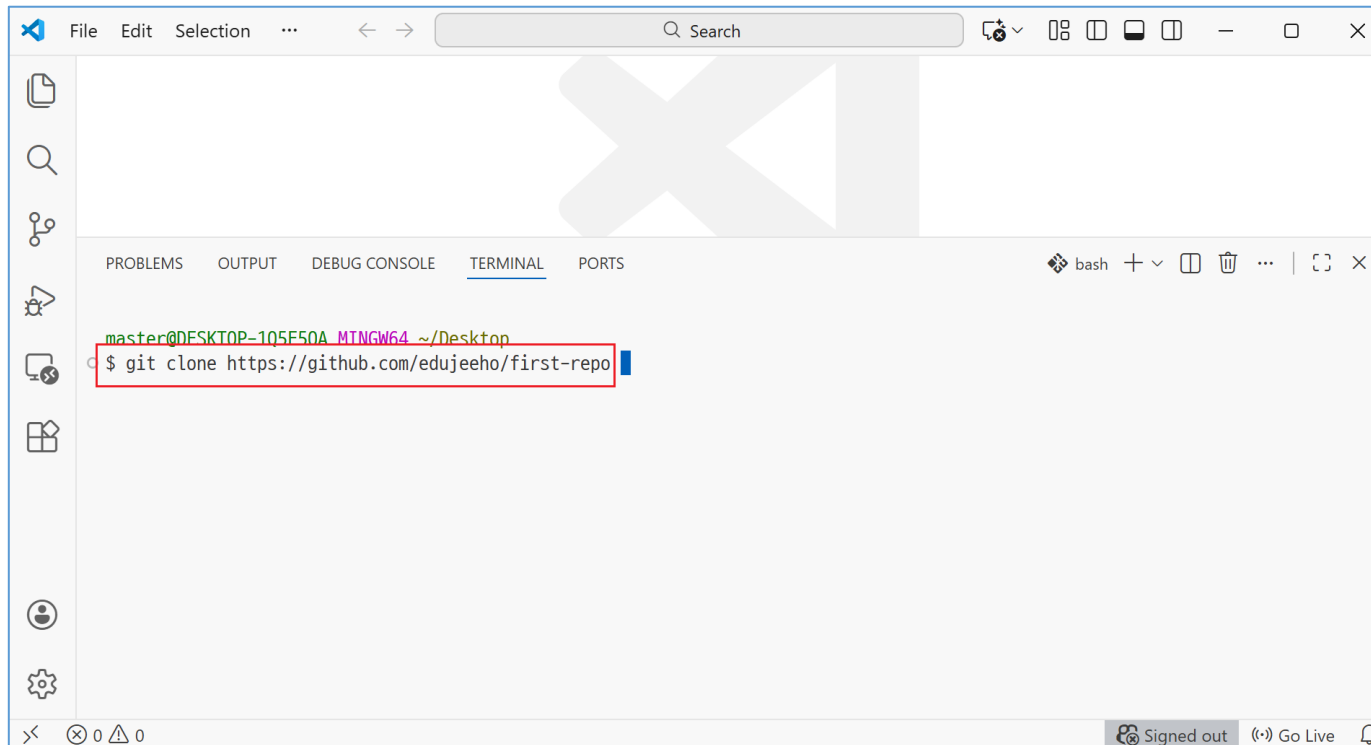
git pull & clone 미리보기 (4/4)

- 원격 저장소의 업데이트 중 변경사항(2의 저장소에만 없는 commit들)만 다운로드



git pull & clone 따라하기 (1/5)

- 바탕화면에서 원격 저장소를 clone 후 폴더명을 git_advanced로 변경

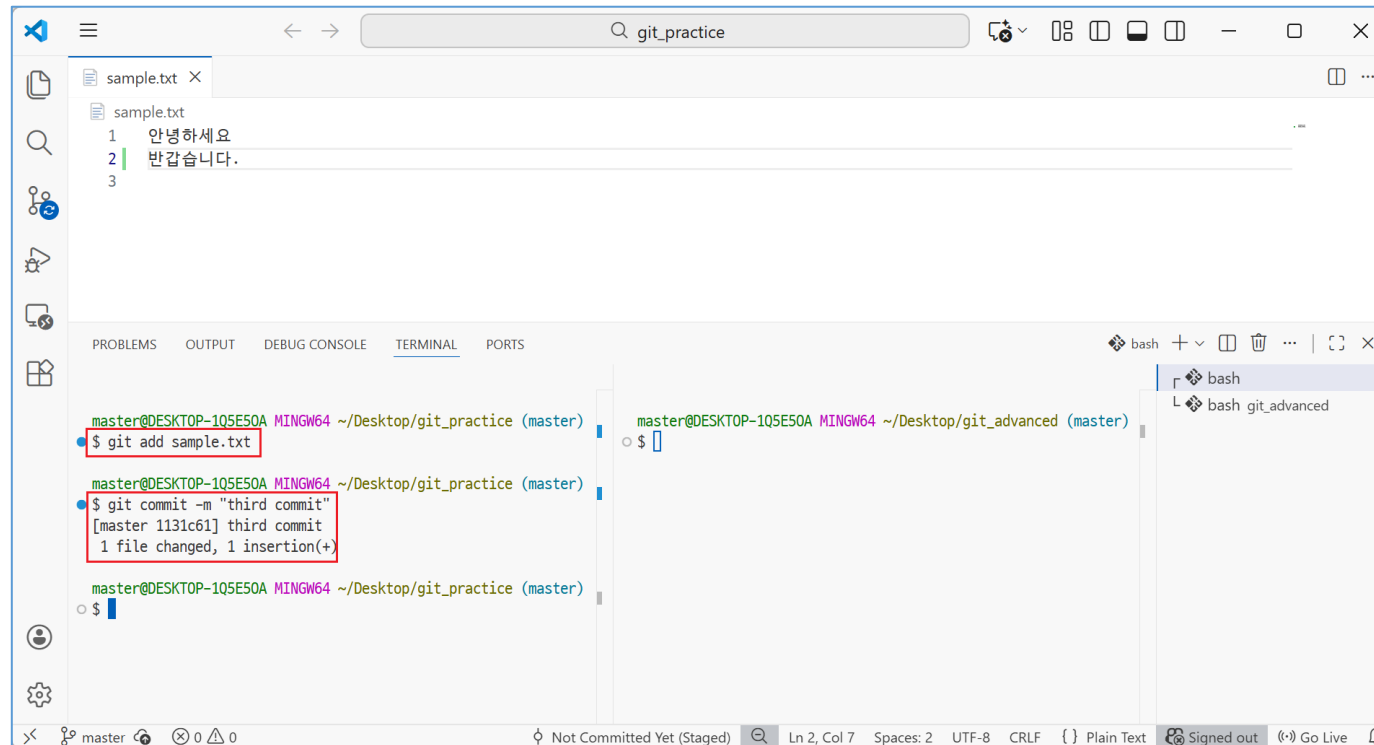


The screenshot shows a Windows terminal window with a light blue border. The title bar includes the VS Code logo and menu items: File, Edit, Selection, and a search bar. The terminal window has a tab labeled 'TERMINAL' and a status bar at the bottom showing 'Signed out' and 'Go Live'. The terminal output shows the prompt 'master@DESKTOP-1Q5F50A MINGW64 ~/Desktop' followed by the command '\$ git clone https://github.com/edujeeho/first-repo' which is highlighted with a red rectangular box.

```
master@DESKTOP-1Q5F50A MINGW64 ~/Desktop
$ git clone https://github.com/edujeeho/first-repo
```

git pull & clone 따라하기 (2/5)

- 기존 git_practice 폴더에서 추가 변경사항 작업 후 commit 생성



The screenshot shows the Visual Studio Code interface with a file explorer on the left showing 'sample.txt'. The main editor displays the content of 'sample.txt' with three lines of Korean text. The bottom panel shows the TERMINAL tab with two terminal windows. The left terminal window shows the execution of 'git add sample.txt' and 'git commit -m "third commit"', with the commit message and hash '[master 1131c61] third commit' and '1 file changed, 1 insertion(+)' highlighted by a red box. The right terminal window shows the prompt 'master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_advanced (master)'.

```
master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_practice (master)
$ git add sample.txt

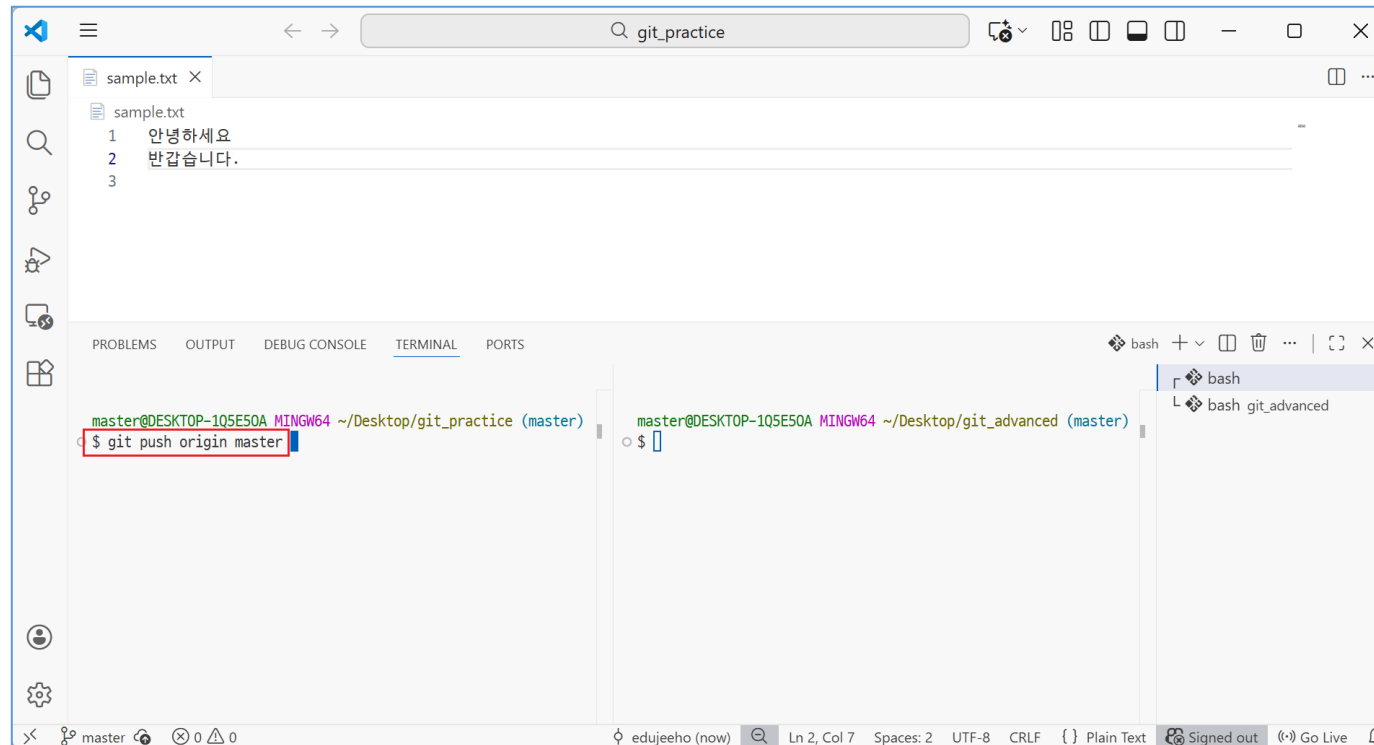
master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_practice (master)
$ git commit -m "third commit"
[master 1131c61] third commit
1 file changed, 1 insertion(+)

master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_practice (master)
$
```

```
master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_advanced (master)
$
```

git pull & clone 따라하기 (3/5)

- 새로운 commit을 원격 저장소에 push



The screenshot shows the Visual Studio Code interface. The top editor pane displays a file named `sample.txt` with the following content:

```
1 안녕하세요  
2 반갑습니다.  
3
```

The bottom panel shows the TERMINAL view with two terminal sessions. The left terminal session is titled `bash` and shows the command `git push origin master` being entered. The right terminal session is titled `bash git_advanced` and shows a prompt `$`.

git pull & clone 따라하기 (4/5)

- git_advanced 폴더에서 업데이트된 변경사항 받기

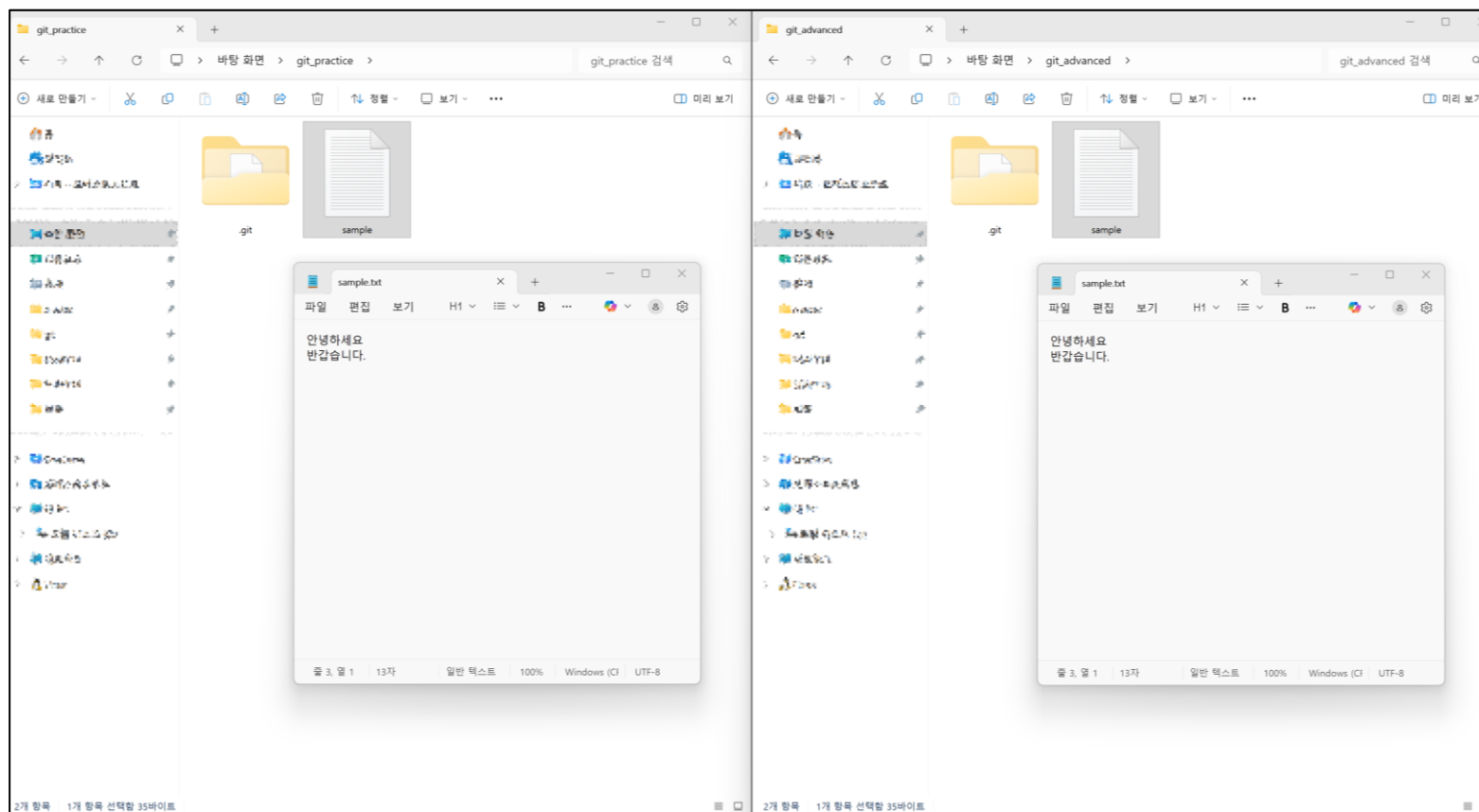
The screenshot shows the Visual Studio Code interface. The top editor pane displays a file named 'sample.txt' with the following content:

```
1 안녕하세요.  
2 반갑습니다.  
3
```

The bottom panel shows the 'TERMINAL' tab with two terminal windows. The left terminal window shows the prompt 'master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_practice (master)' and the command '\$ ' entered. The right terminal window shows the prompt 'master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git_advanced (master)' and the command '\$ git pull' entered, which is highlighted with a red box. The status bar at the bottom indicates the current file is 'Ln 2, Col 7' and the encoding is 'UTF-8'.

git pull & clone 따라하기 (5/5)

- 두 폴더의 파일 모습이 같아졌다는 것을 확인



git 기타 명령어

- `git remote -v`
 - 현재 로컬 저장소에 등록된 원격 저장소 목록 보기
- `git remote rm <remote_name>`
 - 현재 로컬 저장소에 등록된 원격 저장소 삭제
- `git push -set-upstream origin master` (`git push -u origin master`)
 - `master` 브랜치의 원본 브랜치를 `origin`의 `master`를 원본으로 기록
 - 이후 `git pull` 등의 다른 명령어에서 '특별한 인자가 없을 경우' `origin`의 `master` 브랜치의 기록을 가져옴

gitignore



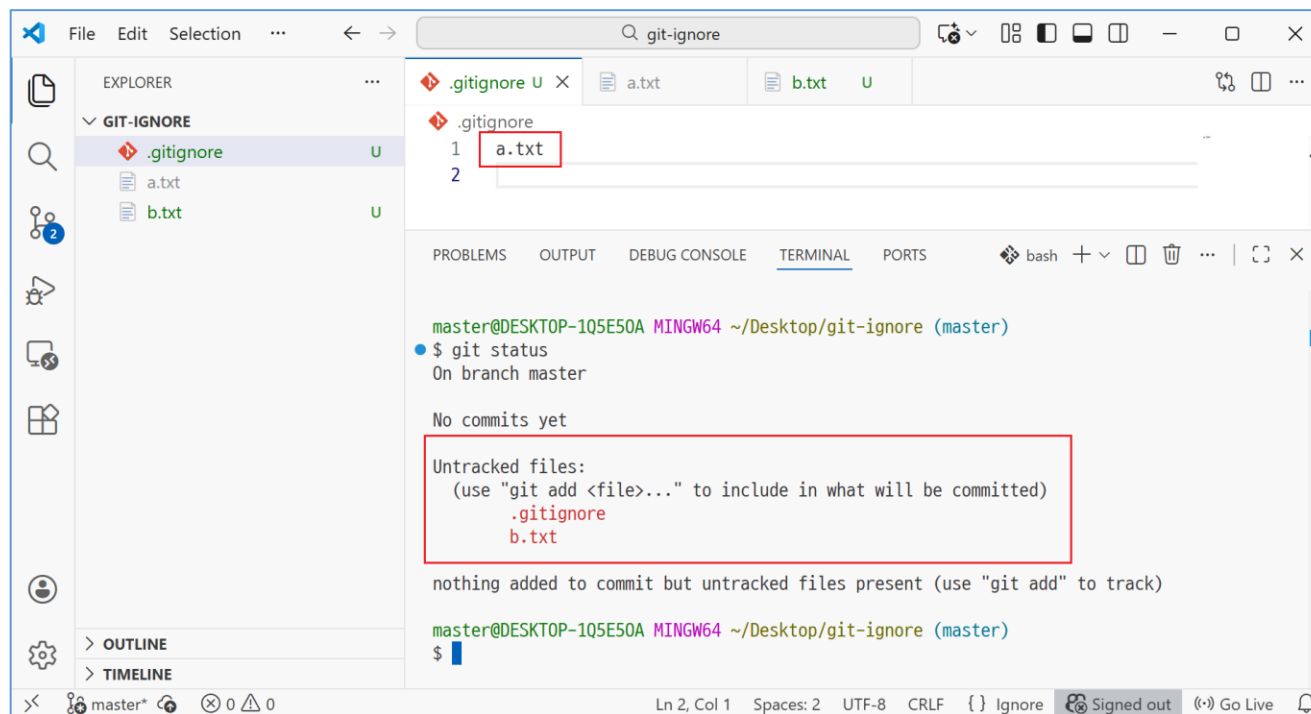
gitignore

Git 에서 특정 파일이나 디렉토리를 추적하지 않도록
설정하는 데 사용되는 텍스트 파일

- 프로젝트에 따라 공유하지 않아야 하는 것들도 존재하기 때문

gitignore 예시

1. .gitignore 파일 생성 (파일명 앞에 '.' 입력, 확장자 없음)
2. a.txt와 b.txt 파일 생성
3. .gitignore에 a.txt 작성
4. git init
5. git status



The screenshot shows the Visual Studio Code interface. In the Explorer pane, the 'GIT-IGNORE' folder is expanded, showing '.gitignore', 'a.txt', and 'b.txt'. The '.gitignore' file is open in the editor, showing 'a.txt' on line 1. The terminal pane shows the output of the 'git status' command:

```
master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git-ignore (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        b.txt

nothing added to commit but untracked files present (use "git add" to track)

master@DESKTOP-1Q5E50A MINGW64 ~/Desktop/git-ignore (master)
$
```

 gitignore 주의사항

gitignore 주의사항

이미 Git의 관리를 받은 이력이 있는 파일이나 디렉토리는
나중에 gitignore에 작성해도 적용되지 않음
(git rm --cached 명령어를 통해 git 캐시에서 삭제 필요)

gitignore 목록 생성 서비스

- 운영체제, 프레임워크, 프로그래밍 언어 등 개발 환경에 따라 gitignore 목록을 만들어주는 사이트
- <https://www.toptal.com/developers/gitignore/>

실습

실습 1 - 원격 저장소에 push 하기

1. 새로운 폴더 생성 후 로컬 저장소 설정
2. commit 목록 생성
3. 새로운 GitHub repository 생성
4. 원격 저장소 추가
5. commit 목록 push

실습 2 - 로컬 저장소 하나에 여러 원격 저장소 추가하기

1. 기존에 이미 origin을 추가한 로컬 저장소에 이어서 진행
2. 새로운 GitHub repository 생성
3. origin이 아닌 다른 이름으로 원격 저장소 추가
4. commit 목록 push



감사합니다

